1. Introduction

1.1 Purpose

This document gives the design of the Pre-Assessment module. The High Level Design (HLD) shows the architecture and components. The Low Level Design (LLD) shows the classes, database, APIs and algorithms needed to code it.

1.2 Module Goal

- Ask the student a questionnaire to estimate learning preferences across the six approved learning pathways.
- Calculate preference scores from self-report and scenario choices; keep objective task performance and response time separate.
- Build an indicative learning-preference profile and identify primary and secondary preferences; do not label these as fixed intelligence types.
- Give the preference profile to Module 2 as an input for recommending content representations.
- Finish in 15 minutes: 20 questions, each with its own time limit

1.3 Design Decisions Made

These are configurable implementation decisions where the requirements do not specify a value. Provisional values must be reviewed during a student pilot; they are not validated measurements.

No.	Decision	Reason
D1	20 questions: 6 self-report, 6 interactive tasks, 8 scenario choices	Provisional short assessment for Grades 6–8. Question mappings must provide reasonable coverage across all six pathways and be checked in a pilot.
D2	Scenario questions map each option to a different intelligence	Scenario options represent different ways to explore the same concept; options are preference choices, not right or wrong answers.
D3	Self-report uses a 5-point scale; response weight = (rating - 1) / 4	Gives a value between 0 and 1
D4	No time penalty is applied to preference scores	Avoid penalizing reading speed or response latency; server timing is used only to enforce assessment limits.
D8	Each question has a time limit; the whole test has a 15-minute limit (900 s)	Provisional duration. A question not answered in time is recorded as timed out and the test moves on; timing must be pilot-tested.
D5	Time is measured on the server	Refresh or tampering cannot reset the timer (PRE-N3, PRE-T4)
D6	Interpersonal and Intrapersonal tasks are situation-based choices with one best answer	These two cannot be tested with puzzles
D7	Teacher/parent observation is optional and separate from the student preference score	Avoid silently mixing observer judgments with student self-reports; enable only after permissions and consent are approved.

2. High Level Design (HLD)

2.1 System Architecture

The module uses a 4-layer architecture: Presentation, API, Service and Data. Each layer talks only to the layer below it.

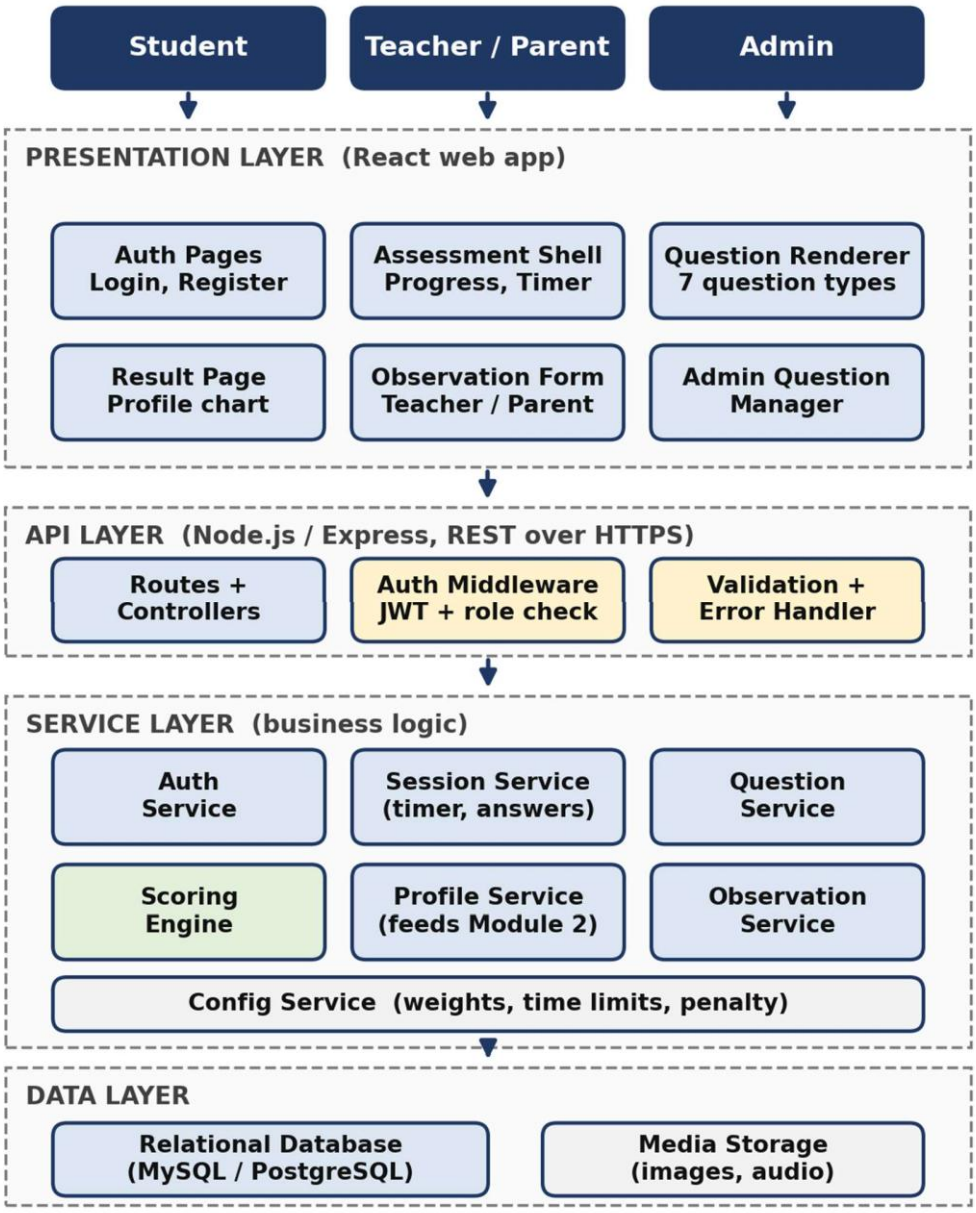


Figure 1: HLD architecture of the Pre-Assessment module

2.2 Component Overview

Layer	Component	Responsibility
Presentation	Auth UI	Registration and login
Presentation	Assessment Shell	Instructions, progress bar, countdown timers (question and total), navigation
Presentation	Question Renderer	Shows approved question types: self-report, scenario choice, and optional interactive tasks.
Presentation	Result Page	Shows indicative learning-preference scores and primary/secondary preferences.
Presentation	Observation Form	Optional teacher observation form; only for authorized users if approved.
Presentation	Admin Question Manager	Add, edit, delete questions and weights
API	Controllers and routes	Receive requests and return responses (REST, JSON)
API	Auth middleware	Checks JWT token and user role

API	Validation and error handler	Checks input and gives clear error messages

Layer	Component	Responsibility
Service	Auth Service	Register, login, password hashing
Service	Session Service	Start or resume session, serve next question, save answers, enforce time limits
Service	Question Service	Select balanced question set; admin CRUD
Service	Scoring Engine	Applies formula, normalises, ranks
Service	Profile Service	Saves the preference profile and provides primary/secondary preferences to Module 2.
Service	Observation Service	Stores optional observations separately from student preference scores.
Service	Config Service	Reads versioned question weights and timing limits; no preference-score time penalty.
Data	Relational database	Users, questions, sessions, responses, scores, profiles
Data	Media storage	Images and audio used in tasks

2.3 Main Data Flow

The sequence below shows one full preference assessment, from start to result. Steps 5 to 10 repeat for each of the 20 questions.

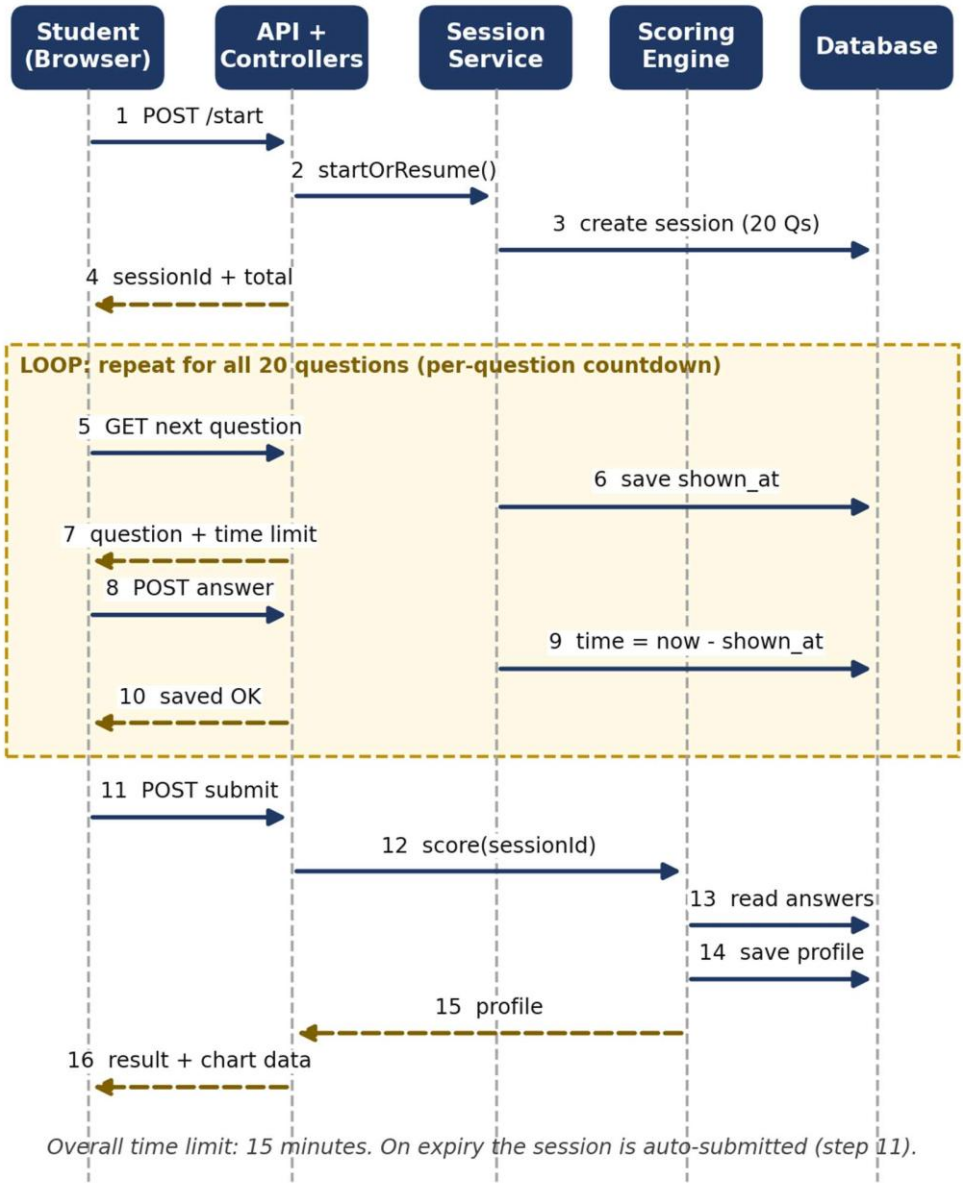


Figure 2: Sequence of one assessment

2.4 Data Design (Overview)

The data design contains the core tables plus scoring configuration. Details are in the LLD.

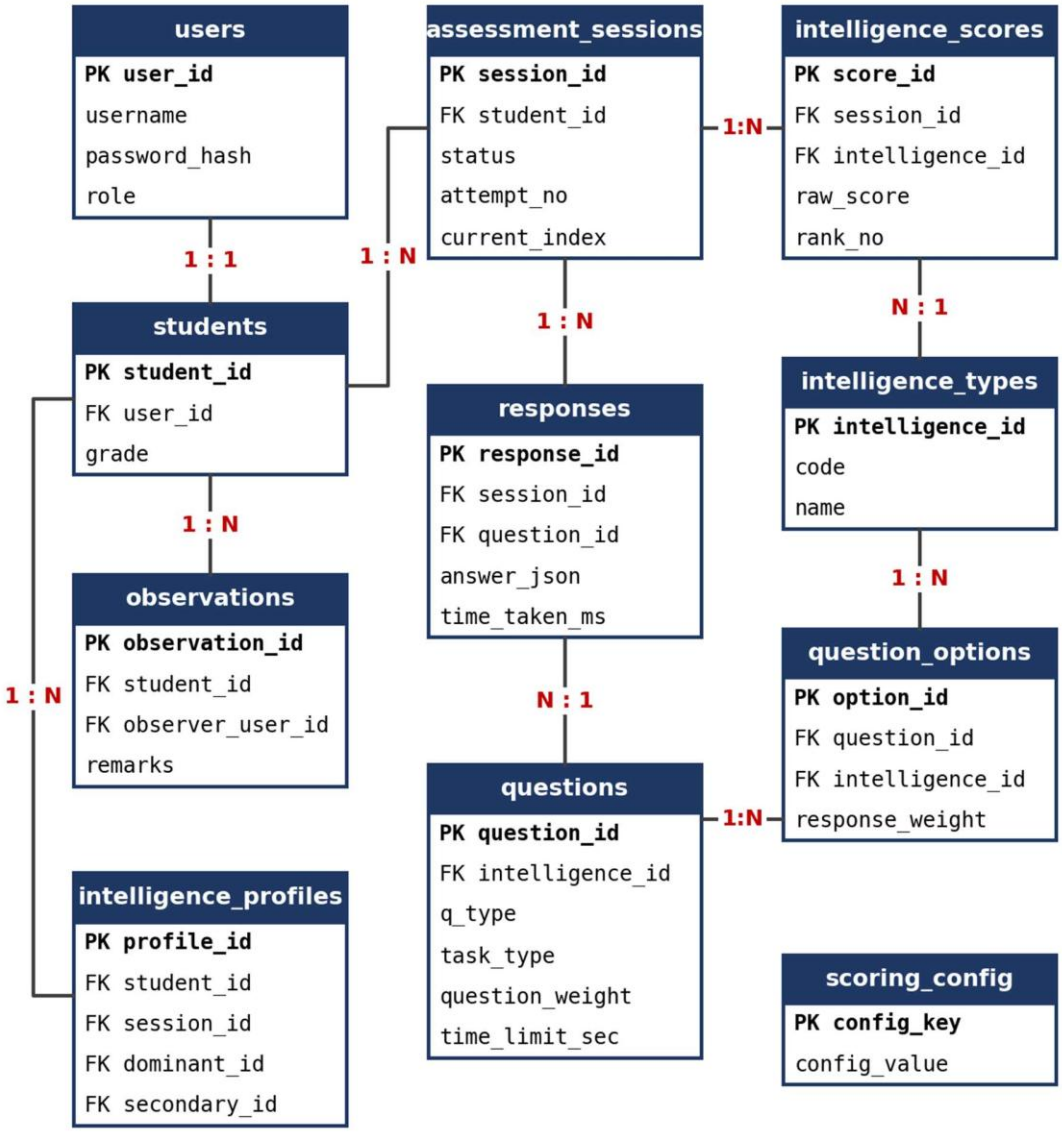


Figure 3: Entity relationship diagram

Only key columns are shown. Full columns are in Section 3.3. Questions and profile scores reference the six approved learning pathways; profiles store primary and secondary preferences.

2.5 External Interfaces

Interface	Direction	Details
Student browser	In / Out	HTTPS REST calls using JSON
Module 2 (Content Learning)	Out	Authenticated profile API returns six pathway preference scores and primary/secondary preferences.
Module 3 (Post-Assessment)	Out	Module 3 receives concept-level pre-test and post-test scores for learning-gain calculation; preference scores are not academic pre-test scores.
Media storage	In	Images and audio for interactive tasks

2.6 Requirement to Design Mapping

Requirement	How the design meets it
PRE-F1, F2 Register, direct to questionnaire	Auth Service; UI redirects when no profile exists
PRE-F3 Question mapped to intelligence	questions.pathway_id or question_options.pathway_id; scenario options can map to preference pathways.
PRE-F4 Question types	Question Renderer with one component per type
PRE-F5 Record answer and time	responses table with shown_at, answered_at, time_taken_ms
PRE-F6, F7 Formula	Scoring Engine (Section 3.6)
PRE-F8 Rank, dominant, secondary	Profile Service ranks six pathways and stores primary/secondary preferences.
PRE-F9 Show chart	Result Page
PRE-F10 Observation	Optional observation service; separate from student preference score.
PRE-F12 Admin edits questions	Admin Question Manager and Question Service
PRE-F13 Re-take	assessment_sessions.attempt_no; only latest profile is_current
PRE-N6 Duration 15 minutes	20 questions; per-question limits add up to 900 seconds; total limit enforced on server
PRE-N3 Timer not reset	Server stores shown_at; client reads remaining time from server
PRE-N4 Save on drop	Every answer saved on submit; session resumes at current_index
PRE-N7 Security	JWT, role checks, hashed passwords, HTTPS

3. Low Level Design (LLD)

3.1 Frontend Components

Component	Props / State	Behaviour
LoginPage, RegisterPage	form fields	Calls auth API; saves token; redirects new student to /assessment
AssessmentPage	session, currentQuestion, index	Controls the flow: start, next, submit
InstructionScreen	none	Explains the test in simple words before starting
ProgressBar	index, total	Shows question number
QuestionTimer	time limit, remaining time from server	Countdown for the question and for the whole test; at 0 the answer is saved as timed out and the next question opens
QuestionRenderer	question	Factory: picks component by q_type and task_type
LikertQuestion	text, scale 1-5	5 buttons (emoji faces); returns rating
ScenarioQuestion	text, options	Option cards (with icons); returns option_id
PatternPuzzle	sequence, choices	Pick the next item; returns choice
ImageMatch	image pairs	Match two sets; returns pairs
RhythmTask	audio_url, pattern	Plays a beat; student repeats by tapping; returns tap times
ClassificationTask	items, groups	Drag items into groups; returns mapping
SequenceTask	steps	Drag steps to the right order (Bodily-Kinesthetic)
ResultPage	profile	Bar/radar chart; shows indicative primary and secondary learning preferences.
ObservationForm	ObservationForm	student, six ratings

Authorized teacher/parent may rate the six pathways if the optional observation workflow is approved.	questions list	Table with add, edit, delete; weight fields

3.2 Backend Classes

Class	Main methods	Notes
AuthController / AuthService	register(), login(), verifyToken()	bcrypt hashing; JWT expiry 2 hours
AssessmentController	start(), getNext(), saveAnswer(), submit()	Thin layer; calls SessionService
SessionService	startOrResume(), nextQuestion(), recordAnswer(), timeoutAnswer(), finish()	Owns session state and timing
QuestionSelector	buildSet(grade)	Picks 20 questions; balances coverage across six pathways and mixes order.
TimerService	markShown(), computeTime(), isExpired()	Server clock only; checks question limit and 15-minute total limit
ScoringEngine	scoreSession(sessionId)	Runs the preference-scoring steps in Section 3.6.
ItemEvaluator (interface)	evaluate(question, answer)	Returns response weight 0 to 1
LikertEvaluator	evaluate()	(rating - 1) / 4
ScenarioEvaluator	evaluate()	Maps the selected scenario option to a preference contribution; no correct/incorrect answer.
TaskEvaluator	evaluate()	If retained, stores task performance separately; correctness does not affect preference scores.
ProfileService	saveProfile(), getCurrent(studentId)	Marks old profile is_current = false
ObservationService	add(), getForStudent()	Stores optional observations separately; does not blend them into preference scores.
ConfigRepository	get(key)	Cached values for the scoring parameters
QuestionRepository, SessionRepository, ResponseRepository, ProfileRepository	CRUD	Only these classes access the database

3.3 Database Schema

users

Column	Type	Constraint
`user_id`	`INT`	PK, auto increment
`name`	`VARCHAR(100)`	NOT NULL
`username`	`VARCHAR(50)`	UNIQUE, NOT NULL
`password_hash`	`VARCHAR(255)`	NOT NULL
`role`	`ENUM(student, teacher, parent, admin)`	NOT NULL
`is_active`	`BOOLEAN`	default true

students

Column	Type	Constraint
`student_id`	`INT`	PK

user_id	INT	FK users, UNIQUE
grade	TINYINT	CHECK 6 to 8
school_name	VARCHAR(100)	optional

pathway_types (6 fixed rows)

pathway_types (6 fixed rows)	Type	Constraint
intelligence_id	SMALLINT	PK
code	VARCHAR(10)	LING, LOGI, SPAT, MUSI, BODY, NATU
name	VARCHAR(30)	Display name

questions

Column	Type	Constraint
question_id	INT	PK
intelligence_id	SMALLINT	FK; NULL for scenario questions
q_type	ENUM(SELF_REPORT, SCENARIO, TASK)	NULL for scenario questions where the selected option supplies the pathway mapping
task_type	ENUM(PATTERN, IMAGE_MATCH, RHYTHM, CLASSIFY, SEQUENCE, WORD, SITUATION)	NULL unless q_type = TASK
question_text	TEXT	NOT NULL
media_url	VARCHAR(255)	optional
question_weight	DECIMAL(3,1)	default 1.0
expected_time_sec	SMALLINT	NOT NULL (timing only; no score penalty)
time_limit_sec	SMALLINT	NOT NULL (hard limit for the question)
grade_min, grade_max	TINYINT	difficulty band
config_json	JSONB	task data (pairs, items, pattern)
is_active	BOOLEAN	default true

question_options

Column	Type	Constraint
option_id	INT	PK
question_id	INT	FK questions
option_text	VARCHAR(255)	NOT NULL
intelligence_id	SMALLINT	FK; set for scenario options
is_correct	BOOLEAN	Used only for objectively scored tasks; scenario preference options have no correct answer
response_weight	DECIMAL(3,2)	0 to 1 preference contribution for mapped options

assessment_sessions

Column	Type	Constraint
session_id	INT	PK
student_id	INT	FK students
status	ENUM(IN_PROGRESS, SUBMITTED, SCORED, EXPIRED)	NOT NULL
attempt_no	TINYINT	1, 2, 3...
question_order	JSONB	list of question_ids in fixed order
current_index	SMALLINT	for resume

started_at, submitted_at	TIMESTAMPTZ	server time

responses (UNIQUE session_id + question_id)

Column	Type	Constraint
response_id	BIGINT	PK
session_id, question_id	INT	FK
option_id	INT	FK, NULL for non-option answers
answer_json	JSON	rating, pairs, taps, order
is_correct	BOOLEAN	NULL for self-report
response_weight	NUMERIC(4,3)	0 to 1
shown_at, answered_at	TIMESTAMPTZ	server time
time_taken_ms	INTEGER	answered_at - shown_at
timed_out	BOOLEAN	true if the limit was reached

intelligence_scores (UNIQUE session_id + intelligence_id)

Column	Type	Constraint
score_id	INT	PK
session_id, intelligence_id	INT, TINYINT	FK
raw_score	DECIMAL(6,3)	Sum of weighted preference contributions only
max_score	DECIMAL(6,3)	Sum of applicable preference question weights
profile_value	DECIMAL(4,3)	Normalized score: raw_score / max_score × 100 (0–100); unavailable if no valid items
rank_no	TINYINT	1 to 6

intelligence_profiles

Column	Type	Constraint
profile_id	INT	PK
student_id, session_id	INT	FK
dominant_id, secondary_id	primary_pathway_id, secondary_pathway_id	FK pathway_types
is_current	BOOLEAN	Enforce one current profile per student transactionally
created_at	TIMESTAMPTZ	

observations and observation_ratings

Table	Columns
observations	observation_id PK, student_id FK, observer_user_id FK, observer_role, remarks, created_at; access authorization required
observation_ratings	observation_id FK, pathway_id FK, rating (1 to 5); unique per observation/pathway

scoring_config

config_key	Default	Meaning
time_penalty	0	No response-time penalty in preference scores
near_tie_margin	Provisional	Difference under this margin is displayed as similar preferences; determine after pilot
observation_weight	0	Observations remain separate; do not blend into student preference score

likert_max	5	Highest rating on the scale
question_count	20	Questions per session
total_time_limit_sec	900	Whole test limit (15 minutes)
min_answered	14	Minimum answered questions (70%) to save a profile

3.4 API Design

Method	Endpoint	Role	Purpose
POST	/api/auth/register	Public	Create student account
POST	/api/auth/login	Public	Returns JWT token
POST	/api/assessment/start	Student	Create or resume session
GET	/api/assessment/{sid}/next	Student	Get next question; saves shown_at
POST	/api/assessment/{sid}/answer	Student	Save one answer
POST	/api/assessment/{sid}/submit	Student	Finish and score
GET	/api/students/{id}/profile	Student, Teacher, Parent, Module 2	Current profile
POST	/api/students/{id}/observations	Teacher, Parent	Add observation ratings
GET / POST	/api/admin/questions	Admin	List and create questions
PUT / DELETE	/api/admin/questions/{qid}	Admin	Edit and deactivate question
GET / PUT	/api/admin/config	Admin	Read and update scoring parameters

Sample: GET next question (response)

```
{
  "questionId": 21,  "index": 5,  "total": 20,
  "qType": "SELF_REPORT",
  "text": "I remember diagrams better than paragraphs.",
  "scale": [1, 2, 3, 4, 5],
  "timeLimitSec": 30,  "testRemainingSec": 640
}
// correct answers and intelligence mapping are never sent to the browser
```

Sample: POST answer (request) and submit (response)

```
POST /api/assessment/12/answer
{ "questionId": 21, "answer": { "rating": 4 } }

POST /api/assessment/12/submit  ->
{
  "dominant":  { "id": 3, "name": "Spatial",  "profile": 1.00 },
  "secondary": { "id": 4, "name": "Musical",  "profile": 0.87 },
  "scores": [ { "name": "Spatial", "profile": 1.00, "percent": 86 }, ... ]
}
```

3.5 Question Bank Design

Part	Count	Details
Self-report	6	One per pathway. Example: “I enjoy explaining ideas to friends.”
Interactive tasks	6	One per pathway as an optional separate performance measure; task correctness is not included in preference scoring.
Scenario choice	8	Eight preference scenarios; options represent different ways of exploring a concept. Validate balanced pathway coverage in pilot.
Total	20	Six self-report, six tasks, eight scenarios. Scenario options and question mapping require pilot review.

Intelligence	Interactive task
Linguistic	Word meaning match
Logical-Mathematical	Pattern puzzle

Intelligence	Interactive task
Spatial	Image matching
Musical	Rhythm task: listen and repeat the beat
Bodily-Kinesthetic	Drag-and-drop steps in the right order
Linguistic	Word meaning match (performance recorded separately)
Logical-Mathematical	Pattern puzzle (performance recorded separately)
Naturalistic	Classify animals and plants into groups

Scenario questions and the learning pathways their options map to (coverage must be balanced and pilot-tested):

Scenario	Option 1	Option 2	Option 3	Option 4
S1	Linguistic (story)	Spatial (diagram)	Musical (song)	Bodily (activity)
S2	Logical (experiment)	Interpersonal (group task)	Intrapersonal (reflection)	Naturalistic (field study)
S3	Linguistic (debate)	Logical (puzzle)	Musical (rhyme)	Naturalistic (observation)
S4	Spatial (animation)	Bodily (simulation)	Interpersonal (role-play)	Intrapersonal (journal)

Rules: cover all six pathways; use language suitable for Grades 6 to 8; do not copy text from reference websites; every question must be reviewed by a teacher. The profile is indicative and must be checked in a pilot.

Sample question record (task type)

```
{ "question_id": 40, "intelligence_id": 4, "q_type": "TASK", "task_type": "RHYTHM",
  "question_text": "Listen to the beat and tap the same pattern.",
  "question_weight": 1.0, "expected_time_sec": 30, "time_limit_sec": 60,
  "config_json": { "audio": "beat_03.mp3", "pattern_ms": [0, 500, 1000, 1250] } }
```

3.6 Scoring Algorithm

3.6.1 Parameters

Symbol	Meaning
rw	Response weight (0 to 1) from the evaluator
qw	Question weight (default 1; 1 to 3)
t, t_exp, t_lim	Time taken, expected time and time limit of the question (t is never above t_lim)
tp	Time penalty = qw x min(penalty_cap, penalty_rate x max(0, t - t_exp) / t_exp)
item score	max(0, rw x qw - tp)
S_i	Raw score of intelligence i = sum of item scores
M_i	Maximum possible score of i = sum of qw of questions where i can score
profile_i	S_i / max(S) (as given in the notes; top intelligence = 1.00)
percent_i	S_i / M_i x 100 (absolute strength, shown to the student)

3.6.2 Steps

```
scoreSession(sessionId):
  if answeredCount < cfg.min_answered: return NOT_ENOUGH_ANSWERS  // ask to re-take
  responses = ResponseRepository.getBySession(sessionId)
  cfg = ConfigRepository.all()
  for each response r:
     q  = QuestionRepository.get(r.question_id)
     if r.timed_out: rw = 0 ; t = q.time_limit
     else: rw = Evaluator(q.q_type).evaluate(q, r.answer)  // 0..1
     over = max(0, t - q.expected_time) / q.expected_time
     tp = q.weight * min(cfg.penalty_cap, cfg.penalty_rate * over)
```

```
    item = max(0, rw * q.weight - tp)
    target = q.intelligence_id  OR  chosen option's intelligence_id
    S[target] += item
for each intelligence i:
    M[i] = sum of q.weight where i can score
    if observation exists: S[i] = S[i]*(1-w) + obsNorm[i]*M[i]*w
top = max(S)
for each i: profile[i] = S[i] / top ;  percent[i] = S[i] / M[i] * 100
ranked = sort(i by profile desc, tie-break below)
dominant = ranked[0] ;  secondary = ranked[1]
save intelligence_scores, intelligence_profiles (is_current = true)
```

3.6.3 Tie-Break Rule

- 1st: higher normalized preference score
- 2nd: if scores are within the configured near-tie margin, show both as similar preferences
- 3rd: use stable display ordering only; do not claim that the tie-break establishes a true preference

3.6.4 Worked Example (3 questions for one intelligence)

Q	rw	qw	t / t_exp	tp	Item score
Q1	1.0	1	8 / 10	0	1.00
Q2	1.0	2	15 / 10	2 x min(0.3, 0.3 x 0.5) = 0.30	2.00 - 0.30 = 1.70
Q3 (Likert 3)	0.5	1	5 / 10	0	0.50
Total		M = 4			S = 3.20 (80%)

Example: if raw preference score S = 3.20 and maximum applicable weight M = 4.00, normalized score = 3.20 / 4.00 × 100 = 80%.

3.7 Time Limit Design

Question type	Count	Expected time	Time limit	Total
Self-report	8	15 s	30 s	240 s
Scenario choice	4	25 s	45 s	180 s
Interactive task	8	30 s	60 s	480 s
Whole test	20			900 s = 15 minutes

- Each question shows a countdown; the whole test shows a 15-minute countdown at the top.
- When a question limit ends, the response is recorded as timed out and the next question opens; it does not receive a preference-score time penalty.
- When the 15-minute limit ends, the test is auto-submitted; unanswered questions are recorded as missing/timed out.
- Both limits are checked on the server; the browser timer is only for display.
- A 2-second grace time is allowed for network delay.
- No time penalty is applied to preference scores; response time is used only for server-side timing and optional operational analysis.
- If fewer than 14 questions (70%) are answered, no profile is saved and the student is asked to retake; validate this threshold during the pilot.
- The instruction screen explains the limits and shows one sample question (not scored).

3.8 Session State and Timer Logic

State	Meaning	Next state
IN_PROGRESS	Student is answering; current_index saved	SUBMITTED, EXPIRED
SUBMITTED	All answers received; waiting to score	SCORED
SCORED	Profile saved	none (a new attempt makes a new session)

EXPIRED	No activity for 24 hours	student starts a new session

- getNext(): if the current question already has shown_at, return it with the same shown_at (no reset on refresh)
- saveAnswer(): time_taken = server now - shown_at; client time is ignored
- An answer that reaches the server after the time limit (plus grace time) is recorded as timed out.
- Answers can be saved only once per question; the next question cannot be skipped

3.9 Validation and Error Handling

Case	Check	Response
Token missing or expired	Auth middleware	401 Unauthorized
Wrong role	Role check	403 Forbidden
Answer for wrong question	question_id must equal current question	400 with message
Rating outside 1 to 5	Range check	400 with message
Duplicate answer	UNIQUE (session, question)	409 Conflict
Answer after time limit	time_taken > time_limit + grace	Saved as timed out; next question returned
Submit before all answered	Count responses = total (unless auto-submit)	400: unanswered questions
Database failure	Try / catch + transaction	500 with generic message; nothing half saved

3.10 Security

- Passwords hashed with bcrypt; HTTPS only
- JWT token with role and resource-level authorization; students can read only their own data; teachers can access only assigned students.
- Correct answers for objective tasks stay on the server; preference mappings are not treated as correct answers.
- Parameterised queries to prevent SQL injection; input validated on server
- Minimize student data; obtain required consent and define retention/deletion rules before using real student data.

3.11 Test Cases

ID	Test	Expected result
TC1	New student registers	Redirected to instructions, then questionnaire
TC2	Refresh on question 10	Same question shown; timer continues
TC3	Close browser, log in again	Resumes at saved question
TC4	Answer with rating 6 (API)	400 error
TC5	Answer slowly but within the limit	No response-time penalty is applied to preference scores.
TC11	Do not answer one question for its full limit	Saved as timed out, score 0, next question opens
TC12	Reach the 15-minute total limit	Auto-submitted; unanswered score 0
TC13	Answer only 10 questions	No profile saved; student asked to re-take
TC14	Change browser clock	No effect; server time is used
TC6	All Musical items correct and fast	Musical is rank 1, profile = 1.00
TC7	Two nearly equal pathway scores	Near-tie rule displays similar preferences; no intelligence claim.
TC8	Re-take assessment	New session; old profile is_current = false.
TC9	Submit with unanswered questions	400 error
TC10	Module 2 calls profile API	Returns six pathway preference scores and primary/secondary preferences; ownership

		is checked.